An intelligent IoT-based Data Analytics for Freshwater Recirculating Aquaculture System

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***Abstract*—Smart farming is essential for a nation whose economy largely depends on agro products. In the last few years, rapid urbanization and deforestation have impacted the farmers. Due to the lack of rainwater harvesting and changing weather patterns, many crop failure cases have been registered in the last few years. To prevent loss of annual crop production, many researchers propose the technology-driven smart farming method. Smart farming is a technology-driven control environment for monitoring and maintaining the crop. Smart farming increases crop production and provides an alternative source of income to small farmers. To promote smart farming in India, the government initiated many pilot projects for integrated aquaculture farming. However, the lack of technological intervention and skill-oriented process makes it difficult for most farmers to succeed in this business. In this paper, we have proposed an intelligent IoT-based freshwater recirculating aquaculture system. The proposed system has integrated sensors and actuators. The sensor system monitors the water parameters, and actuators maintain the aquaculture environment. An intelligent data analytics algorithm played a significant role in monitoring and maintaining the freshwater aquaculture environment. The analytics derived the relationship between the water parameters and identified the relative change. From the experimental evaluation, we have identified that the M5 model tree algorithm has the highest accuracy for monitoring the relative change in water parameters.**

***Index Terms*—Aquacalture, IoT, RAS, edge computing, fog computing.**

I. Introduction

In recent years, we have seen a significant transformation in regular farming. Many urban cities are utilizing their building spaces for intelligent farming. Smart farming is a technology-driven control for monitoring and maintaining the proper growth environment [1]. However, smart farming is still tricky for many unskilled farmers [2]. In India, 70 % population directly or indirectly depends on agriculture. Indian agriculture has a significant contribution to the Indian economy [2]. Over the past few years, the changing weather patterns and global warming have significantly impacted annual crop production. A substantial number of farmers are affected by the loss of annual crop production. To cut short their loss and provide an alternative source of income, the Indian government initiated many pilot projects to promote integrated aquaculture-based farming. India has 2.36 million hectares of ponds and tanks, which offers immense opportunities for aquaculture. Aquaculture is the mean of livelihood for 28 million people in India. Unlike other aquaculture, freshwater pearl cultivation is the most profitable business in current scenarios. In 2020, India imported 1.59 trillion rupees worth of pearls, precious and semi-precious stones. To make India self-reliant on pearl production, the Indian government supports the farmers through subsidies and free training programs. Despite government efforts, the annual pearl production has not made significant progress. The lack of technological intervention and skill-oriented manual operation is the primary cause of poor production. In India, freshwater aquaculture-based farming is still operated manually. Aquaculture farming is very new to most farmers in India. Training the farmers and educating them about pearl farming is a time-consuming and skill-oriented job.

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In this paper, we have proposed an intelligent IoT-based freshwater recirculating aquaculture system. The proposed system intelligently monitors and maintains the aquaculture environment. The system has three significant designs: physical, network, and logical. In physical configuration, we have edge devices integrated with sensors and actuators. The sensors monitor the water parameters, and actuators control the habitable aquaculture environment. The communication network between the physical devices manages data and control flow. The network design has an integrated network between the edge, fog, and gateway devices. The gateway operates the VPN servers for secure communication in the public network. At the same time, the network between the edge node and fog node communicates using a socket. In logical design, we have an abstract representation of entities and processes. The logical design included a front-end data visualization and interactive control utility provided through the Danjgo web framework. At the back-end, we have a relational database for managing the data. A data analytics algorithm utilizes the relational data for forecasting the change in the water parameters and control for the actuators. The whole operation for monitoring and management is scheduled through a real-time clock (RTC). The model selected for data analytics is a non-linear decision tree-based model random forest (RF), M5 Model tree, and gradient boosting machine (GBM).

The proposed system has the following main contribution..

- An Intelligent IoT driven monitoring and management of freshwater pearl farming.

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- An inexpensive regent replacement-based sensor system designed for ammonia testing.
- A relational data analytics to forecast problem, sensor fault detection, and inexpensive monitoring and management.

The remaining paper is organized as follows: Section II presents the related work. Section III presents the baseline data. Section IV present the components of proposed IoT System, Section V, presents the Aquaculture management using data analytics. Section VI presents the experimental results followed by conclusion.

II. Related Work

The rapid expansion of the Internet of Things (IoT) brought ubiquitous networked devices and sensors integration in a variety of intelligent applications [3], [4]. The generated data from these devices requires computational intelligence to extract the knowledge from the data [6]. We have identified a few research works centred around water quality monitoring in the literature review. Most research works focus on water quality monitoring rather than system affordability. Still, technology-driven control farming is pretty expensive to most farmers. We found a few research works centred around data analytical techniques for aquaculture monitoring and management. Using relational data analytics techniques, researchers forecast the change in water parameters relative to some know parameters. This technique keeps the system in a stable state with minimal hardware requirements.

Gao et al. [7] proposed an intelligent IoT-based control and traceability system to forecast and maintain water quality for freshwater fish farming. The whole system is divided into two module, i.e., an intelligent module and tracking module. The intelligent management module includes the integrated sensor assembly (pH, water, temperature, DO, turbidity), data acquisition, data analysis, and database management. The tracking module includes the data visualization, chart and data presentations. The intelligent module process the fish pond monitoring data and forecast the change in the water parameters. The change in water parameters are controlled through various fish-pond farming devices (water pump, aerator, and filters). The collected fish pond sensor data processed using the model tree algorithm. The forecasted change in water quality parameters allows the farmers to monitor and manually control through actuator. Gao et al. [7] proposed system monitors the water quality parameters and visualize to farmers for manual control.

Zhu et al. [8] proposed a remote wireless system for monitoring the water quality in intensive fish culture. The proposed system utilizes the artificial neural networks (ANNs) for predictive analyses of water quality using historical data. Zhu et al. [4] system also control the water quality parameters in time to reduce catastrophic losses. The remote water quality monitoring service offered through CDMA with IPsec-based virtual private network (VPN). The system intelligent module utilize the water temperature, PH, and salinity relative parameters to analyse the temporal trend of dissolve oxygen for an half and hour later. Zhu et al. [8] system address the measurement and control of water parameters in future work.

Simbeye et al. [9] implemented an aquaculture monitoring system based on a wireless sensor network (WSN). The sensor node integrated with water temperature, dissolve oxygen, PH, water level and ZigBee communication. The water quality data gathered and send via a gateway to local server. The local server visualize the data at GUI interface of LabWindows/CVI software platform. The research primarily focused on power management and network system efficiency for monitoring the aquaculture. Simbeye et al. [9] system designed for aquaculture monitoring in personal area network.

Dabrowski et al. [10] proposed a machine learning approach to predict the dissolved oxygen (DO) using pH and water temperature. The data collected from the aquaculture prawn ponds used for this study. Dabrowski et.al [10] analyse the accuracy of DO prediction using LSTM, Linear regression, ANN, and LDs. The experiment results show that the LSTM algorithm produces the optimum results with minimum normalized root mean squared error (NRMSE). Dabrowski et. al.[10] research shows the accuracy of machine learning approach for water quality forecast.

Chen et al. [11] proposed system integrated with water quality sensor and ZigBee based wireless communication. The sensor data gathered from the fish-pond and delivered to the central processing system designed with low power microcontroller. The processed data transferred to Wi-Fi interface to terminal device. The end user can control the entire farm environment through terminal device. The data visualization GUI designed using android platform. Chen et. al. [11] proposed system works in private network with local processing and control. Mahfuz et al. [12] proposed an intelligent microcontroller-based aquaculture monitoring system. A user friendly mobile app visualize the gathered data of the sensor. The whole system connected to solar system for uninterrupted power supply. The change in any water parameters get notified through simple text messages. The proposed system need manual control the change in water parameters for aquaculture management.

Vernandhes et al. [13] proposed an IoT-based intelligent aquaponics monitoring and control system. The end-user has remote access to each physical component via an application. The proposed system consists of sensors, actuator, relay, Ethernet, and router. The author has considered the parameter like temperature, humidity, and soil for this intelligent aquaponics system. The aquaphonic system integrated actuators and sensor maintain the control environment for better growth of the plant. The proposed system in future plan to extend for indoor farming of other crops.

The work in [14] used the multivariate linear regression (LR) and artificial neural network (ANN) to forecast the DO relative to pH, water temperature, and electrical conductivity. Similar work extend in [10] and shown the comparative performance of ANN and LR with support vector machine (SVM). All the competing model have shown a significant accurate forecast of DO. However, these model accuracy of forecast relies on short interval high amount of historical data. Currently, we have identified various ANN models such as LSTM and Deep belief networks are considered to be an appropriate to solve the forecast problem [15][16].

In recent years, some research investigated integrated water quality monitoring and management system based on culture knowledge model and forecasting model [17-19]. However, there system not aimed at the present need to develop fully automated aquaculture. Moreover, there installation cannot achieve the real-time exchange of data and control, which is affecting the water quality and aquaculture production.

In this paper, the proposed IoT system managing the freshwater aquaculture with real-time exchange of data and control. The proposed system designed for freshwater pearl farming where culture breed has specific water quality and management requirement. However, the proposed system dynamic to adjust for other freshwater culture breed.

III. Baseline Data

We have collected the baseline data from the Central Institute of Freshwater Aquaculture Bhubaneswar (CIFA Bhubaneswar). According to the Central Institute of Freshwater Aquaculture Bhubaneswar, the ideal environment for freshwater pearl farming is given in Table. I. For freshwater aquaculture, the water parameters alkaline and hardness must be monitored first. The regional water bodies in different states have different alkalinity and hardness. Alkalinity is the indicator of resistance to acidification. Water bodies that have good alkalinity can prevent the sudden change in Ph. Alkalinity and hardness are closely related since total hardness is associated with calcium and divalent magnesium cations. According to CIFA, the alkalinity of 100 PPM and hardness of 60 PPM are ideal for freshwater pearl farming. From the studies, it is found that hard water is most productive than soft water. Freshwater mussels need soft water for better production. The alkalinity is a time-variant and has a relation with Ph and temperature. In the early morning, Ph is high with moderate or high alkalinity. All biological and chemical reaction has an association with temperature. The warm water resists the solubility of atmospheric dissolved oxygen. The mussels in warm water need more dissolved oxygen compared to cloud water. Similarly, the DO has its prime importance for judging water quality. The pearl farming mussels need an adequate amount of oxygen within the water. The primary source of oxygen in water is obtained from the atmosphere and photosynthesis of plants. The ambient atmospheric oxygen diffusion is significantly less in still water. Similarly, it also diffuses less at high temperatures or in cloudy weather. We directly diffuse the atmospheric air through an oxygen pump and agitation to supplement the required DO in water. Another source of DO in water is through the photosynthesis of plants. In freshwater pearl farming, mussels eat plankton algae. The uneaten plankton algae also influence the water environment. In the daytime, it consumes carbon dioxide and releases oxygen. However, the dissolved oxygen level rises in daylight and reduces at night and on very cloudy days. At night and on a very cloudy day, algae consume oxygen for respiration. Therefore, the excessive growth of algae increases the biological oxygen demand (BOD). The BOD is increased by mixing the ample nutrients in the pond. It also has a relation to time and temperature. During the night, the plankton algae consume the dissolved oxygen rapidly. The plankton density has a relation with dissolved oxygen and Ph. However, the Ph also has a relation to total alkalinity. The fluctuations in pH are less common in higher alkalinity. Similarly, carbon dioxide also has an ill impact on mussels. The higher concentration of carbon dioxide fluctuates the pH. However, the rate of change in pH is relatively less at higher total alkalinity. Carbon dioxide (CO2) concentration increases due to lack of photosynthesis and die-off of phytoplankton. There is a significant need to monitor the pond's dissolved oxygen concentration to minimize the impact of carbon dioxide. Similarly, ammonia concentration is highly lethal for aquatic animals. Ammonia has a relation with dissolved oxygen, pH, and carbon dioxide. The ammonia increases, often decreasing the DO and increasing CO2. Ammonia toxicity is higher at higher pH. The primary source of ammonia is the excreted product of aquatic animals and the decomposition of organic matters. Ammonia is the most crucial parameter to be controlled for better growth of aquatics animals. Table II summarizes some of the recent available literature on the topic. Although some intelligent aquaculture systems have been proposed in the literature, more work needs to be done to integrate the factors that accurately determine the water quality,fault behavior analysis, optimal schedule for feeding, and fault-tolerant monitoring and management. Moreover, the existing systems cannot analyze and interpret real-time data rather monitor through some web application. Still, much work needs to address for real-time control of the system parameters and intelligent data analytics integration at the edge devices or fog devices. As a result, the gap addressed in this paper is relevant to the study.

TABLE I
IDEAL WATER PARAMETERS FOR FRESHWATER PEARL FARMING

Parameters	Values
pH	7-8
DO	8-10 PPM
Alkalinity	100 PPM
Hardness	60 PPM
Food	Plankton Algae
Temperature	15-30 C

IV. Component of Proposed IoT System for Aquaculture Management

This section presents the proposed intelligent IoT-enabled aquaculture management system for freshwater pearl farming. The proposed comprehensive system is divided into four phases of development.

- Physical Design
- Network Design
- Logical Design
- Intelligent Forecasting Models

In Fig. 1, we have shown the conceptual design of an intelligent IoT-based aquaculture system

A. Physical Design

The physical design resides at the edge of the network. The nodes have sensing, actuation, and communication capability.

TABLE II
EXISTING AQUACULTURE MONITORING SYSTEMS.

References	Description
[19]	An intelligent aquaponics system designed using TDS and Ph sensors to schedule optimal fish feeding. TDS and Ph sensor real-time data is monitored through an android application, and fish feeding is automated.
[20]	This paper proposed an automated feeding decision-making system using water quality parameters DO and Temperature. The author used an adaptive neural fuzzy inference system to present an effective control method.
[21]	This research work has discussed the implication of Amazon WEB Services (AWS) for the aquaculture monitoring system by using MQTT protocol for communication with AWS IoT core
[22]	This research has proposed a pH control system based on fuzzy logic control in an aquaponic system. A control system has two Ph sensors and two motor pumps. One Ph sensor is kept in an aquarium, and another is placed in hydroponics.
[23]	In this research work, an author presents a control and monitoring system for ammonia levels for fish cultivation. The designed system is employed a Ph sensor and the MQ-135 gas sensor. The system can perform controls automatically and manually through a smartphone application.
[24]	In this system, a Web application monitors water quality parameters such as Ph and Temperature in real-time for freshwater fish cultivation. The measurement data is stored in a database.
[25]	In this work, the author studies the changing trends for different water quality parameters and manages the control of various devices equipped with aquarium-based fish aquaculture. This study also presents a deep learning model (DL) that correlates the different parameters and produces accurate predictions on the experimental data set.

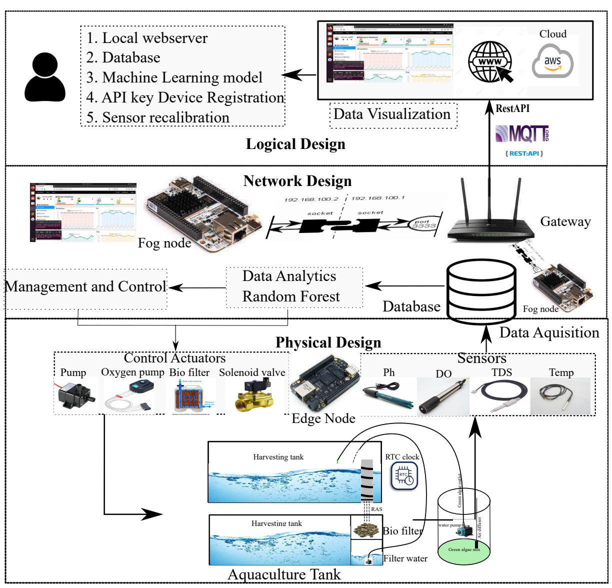

Fig. 1. Conceptual design of intelligent IoT based Aquaculture system

The integrated sensors measure the change in water parameters, and actuators control the change within the threshold. The edge node supports an optimized networking stack that allows the system to access it remotely. The edge nodes in the proposed system are integrated with analogue base dissolved oxygen (DO), Ph, Electric conductivity (EC), RTC timer, and temperature sensors, as shown in Fig. 2. These

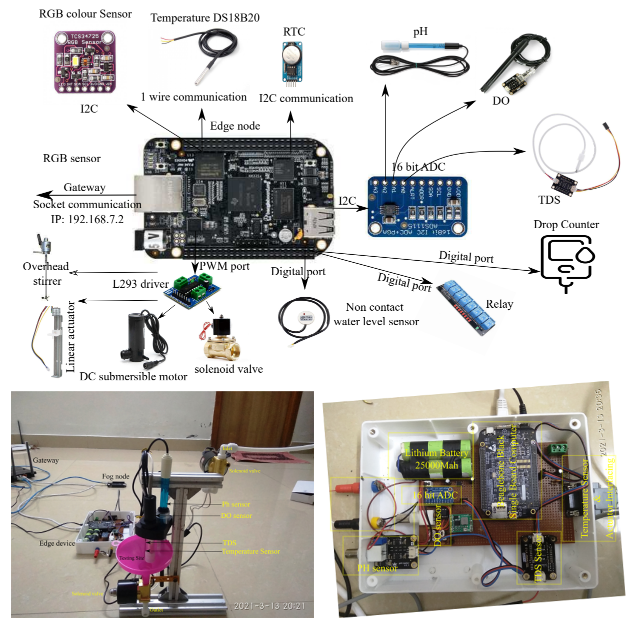

Fig. 2. Edge node integrated sensors and actuators

sensor assemblies are kept outside the aquaculture tanks, as shown in Fig. 3. To accurately measure the change from the sensor data, we have integrated the edge node with ADS1115 16-bit precision analogue to digital converter (ADC) [26]. The ADS1115 ADC provides four input and output analogue channels with a programmable gain amplifier (PGA) module. The high-resolution 16-bit ADC is integrated with DO, Ph, and EC analogue base sensors. Through the I2C interface, ADS1115 ADC communicates the sensor data to the edge node. The 1-wire DS18B20 temperature sensor directly interfaces with the edge node. For edge computation and networking, we have selected the Beagle-bone Black. The Beaglebone black has a variety of general-purpose input-output ports, which allows interaction with various communication interface protocols. The on-board 4 GB internal flash memory provides sufficient space to organize the data in the relational data model. The board has a 1 GHz ARM cortex processor, consuming less power and providing adequate processing capability. To maintain the aquaculture environment, we have used the actuators like oxygen pumps, water pumps, filters, heaters, and solenoid valves. The direct current (DC) operated actuators are connected through the L293 driver. The alternate current (AC) ran actuators are connected through relays. In the proposed system, we have used the sensor and actuators-based collaborative subsystem for aquaculture management. The following subsystem is used to support the automated control for aquaculture management.

1) Test setup
2) Regent replacement based Ammonia Sensor system
3) Power management
4) Automated smart feeder
5) Recirculating Aquaculture System

1) Test Setup: In the proposed system design, we have an edge node equipped with temperature sensors (DS18B20), pH sensor (DFRobot Gravity, model SEN0161), and dissolved oxygen (DFRobot Gravity model no: DFR1628), ammonia sensor, and water electrical conductivity (EC) sensor. Similarly, we equipped the edge node with actuators like water pumps, aerators, feeders, biofilters, water heaters, and solenoid valves. Except for the temperature sensor, all the sensors are

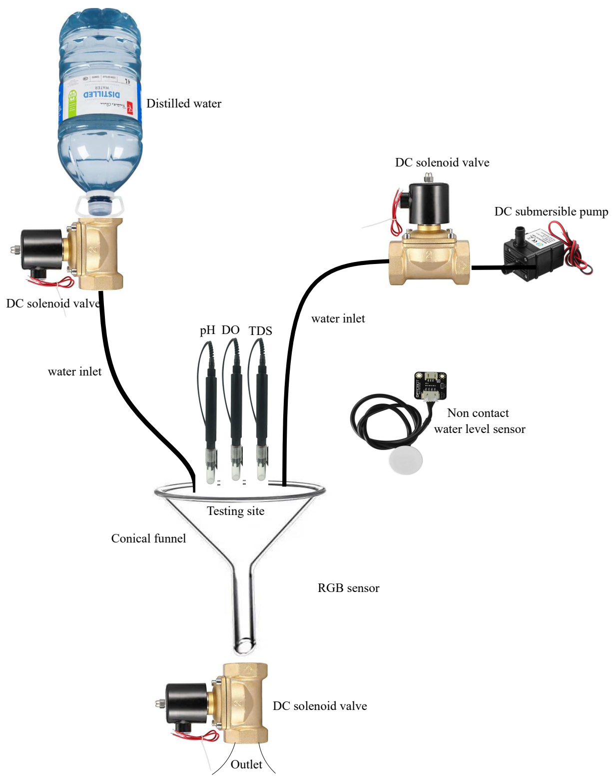

Fig. 3. Test equipment setup

kept outside the main harvesting tank, as shown in Fig. 3. This arrangement prevents the sensor monitoring tip from corrosion and extends their lifetime. The integrated actuator water pump and solenoid valve carry the water from the main harvesting tank to the testing site. After the water quality test, the solenoid valve releases the water from the conical funnel. The non-contact optical water level sensor turns on and off solenoid value and water pump based on the test requirement water level in the funnel. After the end of every test, the distilled water (EC= 10 PPM) pump to clean the sensor probe. This cleaning is necessary for accurate measurement and sensor lifetime. The water quality test is scheduled through RTC. From the experimental study, we have found that water quality is not so sudden a change in the culture pond. Various factors affect the water quality parameters, such as water temperature, weather, feeding, and excreted by-product. The water parameters are inconsistent many times in a day. From the observation of the month of the recorded data, we have identified the best schedule for sensing and actuation of the freshwater pearl aquaculture. However, the system is capable enough to transform into any schedule or continuous monitoring and management.

2) Regent replacement based Ammonia Sensor system: Ammonia is directly toxic to culture breeds in the unionized form, which is favoured at high temperatures and pH. It also reduces the ability of culture breeds to utilize oxygen. Ammonia gets introduced into the water through excreted metabolic waste and the decomposition of organic matter. We have designed an inexpensive regent replacement-based ammonia sensor system to automate ammonia testing. The regent replacement-based ammonia sensor system automates manual regent-based ammonia testing. In the regent replacement-based ammonia sensor system, we have used the water level sensor, drop counter and colour sensor. Similarly, we have used the actuators like water pump, overhead steerer, check valve, and solenoid valve, as shown in Fig. 4. The scheduled set for water

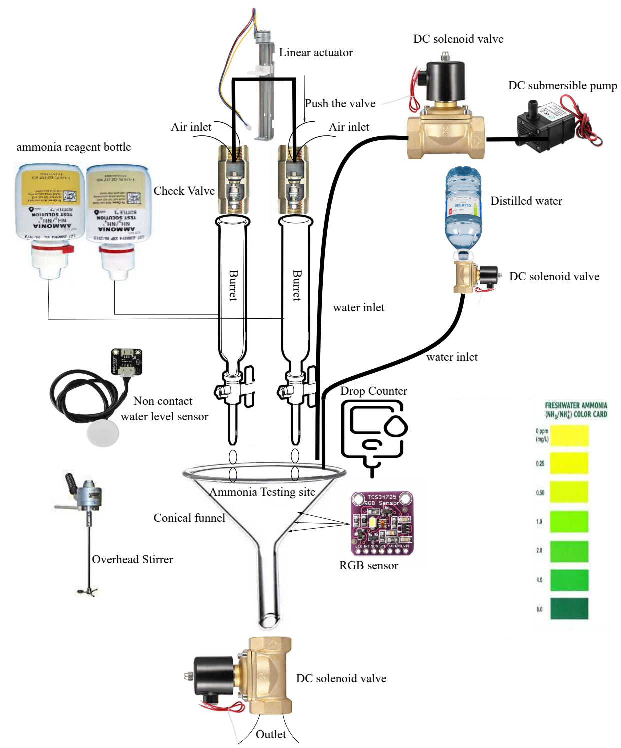

Fig. 4. Regent Replacement based Ammonia Sensor system

quality monitoring activates the ammonia sensor system. At the scheduled time, the edge not integrated DC submersible pump and solenoid valve start their operation to carry the water for quality test. The non-contact water level sensor immediately closes the solenoid valve and pumps when the required amount of water is in the conical funnel. Ammonia testing regent is available in the burette, which is controlled through a check valve. The overhead stepper motor pushes the check valve to make an air entry, resulting in ammonia regent drop released into the conical funnel. To manage the required amount of regent drop, we have a drop counter which signals to the stepper motor and closes the check valve. The drop counter is designed through an optical QRD 114 proximity sensor, invert Schmitt trigger (SN74HC14N) and OR gate (MN4072B IC). The regent in conical funnel mix with overhead stirrer motor. The test result is recorded through an RGB colour sensor and translated into the corresponding numeric PPM value. The test water in the conical funnel is released through the solenoid valve. After the test experiment,

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we pump the distilled water to clean the testing site for the next test. The filtered water is adequately mixed with the overhead stirrer motor and released from the solenoid valve.

3) Power System: The aquaculture system needs regular monitoring and management. In the power subsystem design, we have used the 160 watts, 12-volt monocrystalline solar panel, 24-volt, 30 Amp solar controller, 12-volt lead-acid batteries, and 12-volt relay, 200 Watt AC inverter, and 12-volt DC-DC buck converter for an uninterrupted power supply, as shown in Fig. 5. The proposed aquaculture monitoring and management are scheduled through RTC lock. The missing schedule may affect aquaculture productivity. The water parameters are inconsistent and fluctuate many times in a day. Therefore missing a schedule monitoring and management may impact aquaculture productivity. The proposed power subsystem operates the DC and AC devices. A DC-DC buck converter supplies the required power to the DC-operated overhead stirrer, solenoid valve, and stepper motor. Similarly, the AC inverter operates the 20-watt submersible pump, 15-watt air pump, and 50-watt water heater.

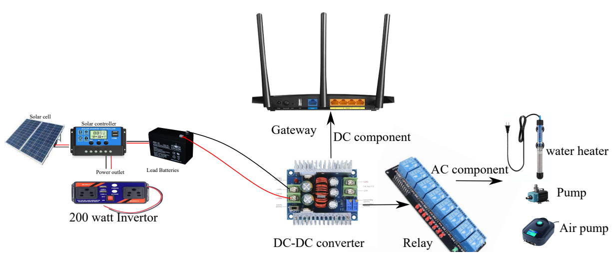

Fig. 5. Power subsystem

4) Automated Smart Feeder: In the aquaculture system, we need proper feed monitoring of the mussels. The uneaten feed has various consequences on the health of the mussels. Overfeeding leads to the loss of lives and increases the water's food pollutants [15]. The overfeeding also increases the excreted by-product in water, which increases the concentration of toxic ammonia and nitrate. The ammonia and nitrate are lithal for mussels. During night uneaten algae consumes the dissolved oxygen and introduces CO_2. However, similar algae in the daytime contribute dissolved oxygen as the by-product of photosynthesis. To balance the aquaculture ecosystem, we must

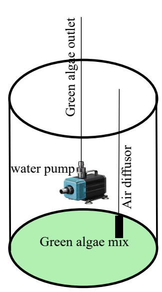

Fig. 6. Automated smart feeder

maintain the proper algae concentration. The overfeeding and residual remaining in water for a longer time could seriously impact water quality. The green algae are soluble in water and easy to feed the mussels. The intelligent feeder system has integrated a submersible pump and agitator, as shown in Fig. 6. The agitators mix the green algae solution, and the submersible pump carries it to the harvesting tank. The smart feeder is scheduled through the RTC clock. The feed time is essential to maintain the water quality parameters. In the daytime feed, the green algae improve the dissolved oxygen concentration as a by-product of photosynthesis. Besides, the feed at night consumes the dissolved oxygen from the harvesting tank. We have a recirculation aquaculture system to regulate the uneaten feed from the harvesting tank. The RAS is scheduled to operate after the overnight feed and whenever there is a significant change in water quality.

5) Recirculating Aquaculture System (RAS): The RAS system filters the culture water and reuses it after treatment. RAS system recirculates the water via biofilters and removes aquatic animal excreted by-products and decomposing waste in the water. Due to a lack of technological intervention, aquaculture farmers find it difficult to do manual monitoring and maintain water quality. In open pond aquaculture, various kinds of waste are mixed with the water. The waste decomposes and influences the water quality. The proposed RAS system has

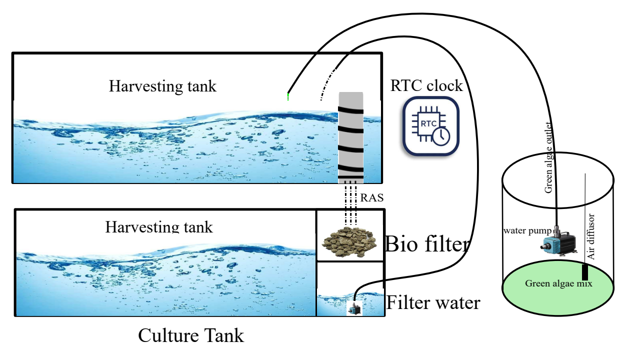

Fig. 7. Recirculating Aquaculture System

water pumps and a Biofilters chamber, as shown in Fig. 7. The RAS system pumps the water from the harvesting tanks to the Biofilter chamber. The filtered water pumps back to the harvesting tank. This process is repeated just after the feed and when there is a significant change in water quality. The proposed RAS solves the central problem of ammonia and balances Ph and DO requirements.

B. Network Design

The proposed network design has a communication network between an edge node, fog nodes, and gateway, as shown in Fig. 8. The socket communication API facilitates sensor data and controls communication between the edge and fog nodes. The gateway is the mediator of this communication between private and public networks. In the proposed network design, fog node placement is essential for data analysis and control. Fog nodes have local databases and analytics algorithms to process the data. The time scheduled monitoring and management are controlled by the fog node. However, the Amazon Web Services (AWS) EC2 infrastructure-as-a-service also runs an identical web server. The data processing

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close to the edge node provides the advantage of keeping the system in the ideal state without missing the schedule monitoring and management. The ability of the current public cloud is insufficient to handle the issues such as latency and bandwidth [31], [32]. If the cloud server manages the entire analysis and control, it can lead to many unwanted results. The connection interruptions between the edge node and cloud servers may lead to the loss of timely scheduled operations at the edge node. We brought the cloud analysis near the edge through fog nodes and avoided scheduled data analysis and control disruption. However, the fog node can only do intermediate data processing and storage. The public cloud manages the scalable storage and processing of sensor data. The web server uses the Rest API for data and control

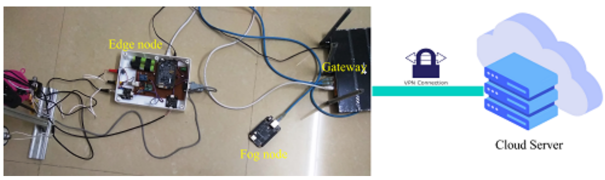

Fig. 8. Network configuration between the nodes

exchange. The data and control are encoded using Base64. We have also used MQTT and RestAPI between the fog node and the public cloud server. The edge node and fog nodes communicate via a gateway to the public cloud, as shown in Fig. 9. We have used the TP-link AC1750 wireless dual-band gigabit router for gateway configuration. The dual-band router that has IEEE 802.11ac/n/a 5 GHz and IEEE 802.11n/b/g 2.4 GHz standards with 450 Mbps and 1300 Mbps bandwidth, respectively. Using the extra router features like virtual private network (VPN) and Network address translation (NAT), we have configured the secure gateway to a public network. The OpenVPN server at the router provides a secure network for control and data exchange in the public network. The VPN client securely allows the remote user to monitor and maintain the aquaculture.

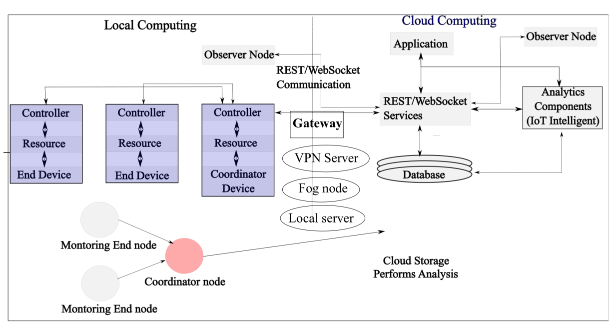

Fig. 9. Network configuration between the nodes

C. Logical Design

In a logical design of an IoT, we refer to the abstract representation of entities and processes. An IoT system comprises various functional blocks that provide capabilities for identification, sensing, actuation, communication, and management. We have used the Django web framework at the front end to monitor and maintain the physical activities. The front-end graphical user interface (GUI) has various control and data visualization options, as shown in Fig. 10. The

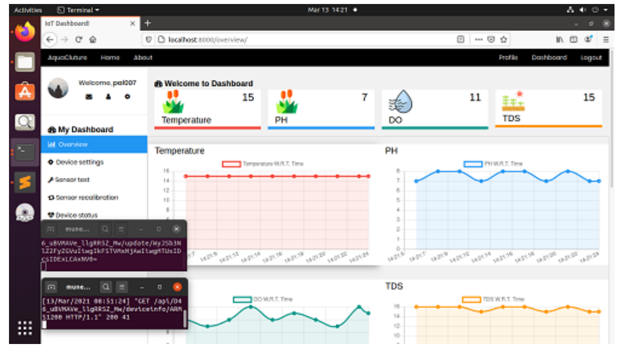

Fig. 10. IoT dashboard

GUI was implemented using HTML5+CSS+Chart.js+JQuery technology stack. We used the Python programming language and MYSQL database for the back end. The web server is deployed in a public network at the cloud instance and in a private network at the fog node. At the cloud server, we have configured two databases. The first database keeps records of all the sensed water parameters, while the second database keeps only the latest data record. The first database recorded data is used for data analytics, while the second database is used for data visualization at the front end of GUI. To simplify the data analytics in the cloud, we have used the docker container-based visualization. The container-based visualization facilitates the critical issues of application developments. The container-based visualizations just need application and associated binaries and libraries. The container isolation is done at the kernel level without requiring any guest operating system, minimizing space and library copies. The isolated data analytics models run in each container, allowing the scale or update of the microservice without impacting other application features, as shown in Fig. 11. It is a swift, low-cost, and elegant isolation framework. The

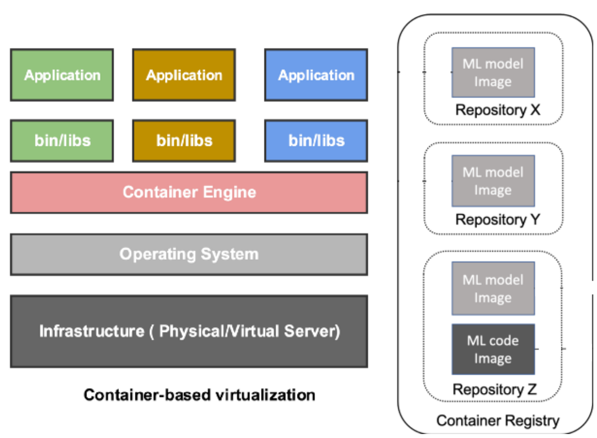

Fig. 11. Container-based visualizations

isolated containers forecast the change in water parameters through different non-linear decision tree-based models. The IoT dashboard has additional features like recalibration, fault detection, and status monitoring of sensors and actuators. The data analytics visualize the forecasted data along with actual data. This feature lets us know the difference between real sensor data and analytics-driven predicted data. To protect each device's identity, we have used a random API key for device registration. The API key is unique to every device in a network. The customer signs up and changes the API key. The remote login to the edge device is protected through a VPN server running the TP-link AC1750 wireless dual-band gigabit gateway router.

D. Intelligent Forecasting Models

To simplify the profitable pearl farming business, we have used the data analytics approach for monitoring and maintaining the aquaculture. The predictive model analyses the change in water parameters and effectively operates the actuators. In this study, we analyse the feasibility of the tree-based model to forecast the water parameter in the presence of a low amount of data. We focus on two main scenarios: 1) estimating water parameters based on the relative change in other parameters. 2) forecast the change in water parameters based on the historical data. Besides, we also compare the performance of the tree-based model with existing multivariate linear regression (LR) and artificial neural network (ANN). The prime focus of this study is to analyse the DO forecast relative to pH, water temperature, and TDS.

The most common approaches for water quality prediction in aquaculture include the following:

- Autoregressive moving average (ARMA).
- Autoregressive integrated moving average (ARIMA).
- Markov model.
- Support vector regression (SVR).

These models are inappropriate for prediction in aquaculture water quality parameters as they only consider the linear relationship. The water quality parameters are inconsistent due to various environmental factors; hence linear prediction models are inefficient in correlating the relationships between multiple predictors and their respective variables. These models also take a long prediction time, making them unsuitable for predicting the non-linear relationship [25], [27]. Deep learning models like long short-term memory (LSTM) may need to be more stable for predicting all real-time dynamics. The predictive processes can be linear to a single target quality parameter and its dynamics over time between multiple predictors and their respective variables. LSTM and gated recurrent unit (GRU) are flexible in capturing the non-linear relationship in water quality parameters [28]. Due to its outstanding results in time series prediction, LSTM is the most popular forecasting DL technique. For time series prediction, LSTM and GRU models could be better at keeping long-term memory, especially for extended sequences. In time series forecasting, historical observations influence the prediction value at the present step. In certain circumstances, the observation step that had a significant impact may have appeared long before the current step. Recent research has shown that the ability of LSTM models to extract information about long-term relationships from historical observations remains a crucial performance constraint. Theoretically, it has been demonstrated that ordinary LSTM lacks long memory from a statistical standpoint [29]. Compared to Naive Bayes, K Nearest Neighbours, and Support Vector Machine, decision tree learning and neural networks result in better and more consistent performance. Known as KIG-ELM, the hybrid DO prediction model combines K-means, enhanced genetic algorithms (IGA), and extreme learning machines (ELM) and is based on edge computing architecture. This model distributes data acquisition, processing, and dissolved oxygen prediction among sensing nodes, routing nodes, and servers. For DO prediction, an extreme learning machine is implemented. Because of the unstable prediction performance constraint, it takes a lot of time to obtain high precision using the multi-scale decomposition method. Multi-parameter methods, which use several related parameters as input and the dissolved oxygen content as an output to forecast future dissolved oxygen, still have some significant issues, such as insufficiently processing DO data effectively and failing to recognize the characteristics of DO content changing. The DO time series data is volatile. During sunrises and sunsets, forecast accuracy typically drops off quickly [30]. In our proposed method, we have compared the accuracy of models like ANN, GBM, RF and M5 model trees. We have chosen the models mentioned above due to certain advantages of these models. Table III summarizes the befits and limitations of the selected models.

To develop the forecast model, we use a model tree (M5) and compare its performance with a gradient boosting machine (GBM), random forest (RF), and artificial neural network (ANN) model. Unlike the linear model, the tree-based models support the non-linear relation; thus, they are suitable for monitoring the actual change in water parameters. The random forest consists of many individual random decision trees that operate as an ensemble. Each random tree serves as a separate predictor based on the decision tree. The low correlation between the individual trees produces ensemble predictions that are more accurate than any individual predictions. We must select the appropriate feature vector rather than random guessing to optimize the RF prediction. Also, the generated unique decision tree should have a low correlation. Similarly, gradient boosting successive improve the performance of the CART algorithm. The algorithm initially assigns an equal weight during the training of a decision tree. After evaluating the initial tree, we modify the weight for a better prediction. The following tree increases the weight for those challenging to classify and reduces the weight that was easy to classify. The subsequent tree provides a better forecast than its predecessor. The final prediction is based on the ensemble weighted sum of the prediction made by the previous trees. Besides, the model tree-based predictive analysis generates a decision tree hierarchy using simple models like linear regression, logistic regression, etc. Initially, the given feature vector is fitted with a linear model. Later, the error is estimated between the actual target and predicted values. The least error features are chosen as the conditional branch of the decision tree. The subsequent split goes until the leaf node has an optimized linear model

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TABLE III
COMPARISON OF FORECASTING MODELS

Machine Learning Models	Benefits	Limitations
Gradient Boosting Machine (GBM)	Build trees one at a time. Each trained tree helps to reduce error. There are typically three parameters - the number of trees, depth of trees and learning rate, and each tree built is generally shallow.	Training generally takes longer because trees are built sequentially. More sensitive to overfitting if data is noisy. Harder to tune than RF.
Random Forest (RF)	Random selection of attributes at each node. Easy to understand and interpret. RF leverages the power of multiple decision trees. More robust and less likely to overfit on the training data. RF is much easier to tune than GBM. There are typically two parameters in RF: the number of trees and the number of features selected at each node.	More than a single tree is required to produce effective results. Need large dataset. A large number of trees may make the algorithm slow for real-time prediction. Biased in favour of those attributes with more levels.
Artificial Neural Network (ANN)	Ability to learn and model non-linear and complex relationships. Handle more than one task at the same time. Loss of one or more cells, or neural networks, influences the performance of Artificial Neural network.	Need processors that support parallel processing. Hardware dependent and sometimes unreliable
M5 Model Tree	M5 model tree can simulate the phenomena with very high dimensionality up to hundreds of attributes. M5 selects the split, which maximizes the expected error reduction.	M5 is a large tree-like structure that may cause overfitting. Consequently, the tree must be pruned back. Work best with limited dataset and hardware. Computational cost grows very quickly when the number of features increases.

fitted on the partial feature vector. The trained model tree effectively forecasts the change in water quality parameters. The recursive optimization of the decision tree may lead to an over-fitting problem. To deal with the over-fitting problem, we have used the pre-pruning method to filter out the anomalies in the dataset using the local outlier factor (LOF). Similarly, we have applied the post-pruning method to prevent the decision tree growth to its full depth. The post-pruning method simplifies the model tree and optimizes the prediction accuracy. Similarly, the ANN models are the conventional approach for time series forecasting. However, unlike the tree-based model, ANN-based models need a larger dataset for training and testing, and it isn't easy to interpret the information from the trained model. We analyse the performance of the multilayer perception ANN model with a ReLu activation function.

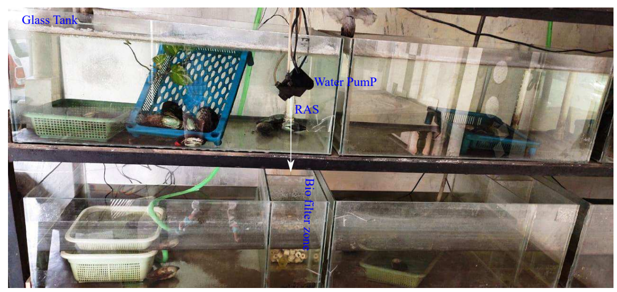

Fig. 12. Aquaculture main harvesting tank

V. EXPERIMENTS AND RESULTS

A. Experimental Setup and Control System

We have used the glass aquarium for freshwater pearl farming. The Lammelidens marginalis mussel species is used for pearl production, as shown in Fig. 12. Each mussel is implanted with two designed, crafted mussel shell nuclei. In the implantation process, each mussel passes through a surgical procedure. After successful implantation, each mussel is kept in post-operating care tanks with antibiotics and an oxygen pump. Few mussels reject the nuclei during the post-surgical operation, and few accept the nuclei. The accepted nuclei mussels are transferred from the post-operative care tank into the main harvesting tank. The mussels need to keep in a controlled environment for better pearl production. Similarly, we equipped the edge node with actuators (pumps, aerators, feeders, biofilters, etc.).In the proposed system, we have an edge node equipped with temperature sensors (DS18B20), pH sensor (DFRobot Gravity, model SEN0161), and dissolved oxygen (DFRobot Gravity model no: DFR1628), and water electrical conductivity sensor (TDS).

The experiential data is collected from the main harvesting tank. We studied temperature's impact on all biological and chemical reactions in the initial data processing. The temperature variation significantly affects water parameters for an entire day. Therefore, we have kept five different timing for data acquisition. The first data acquisition timer was set to 6.00 AM when the temperature and dissolved oxygen remained at their lowest and Ph at their highest point with moderate alkalinity. The second data acquisition time was set at 10 AM when the intelligent feeder system and aeration started operation. The third data acquisition time was set to 3 PM, when temperature, Ph, DO remain at their highest point. The fourth data acquisition timer was set to 10 PM when recirculating aquaculture system started operating. The recirculating water passed through a biofilter to remove the residual plankton feed. The access plankton at night consumes the dissolved oxygen. The analytic algorithm monitors the change in DO and operates the aeration. The fifth data acquisition timer was set to 1 PM when the aeration requirement was analyzed through data analysis. The system operation is monitored and controlled through a real-time clock (RTC). We simulate the system behaviour by changing the water parameters to validate the control system. The intelligent data analytics algorithm schedule the requirement for aquaculture management based on the change in the water parameters.

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B. Data Preprocessing

The edge node converts the raw sensor data into a simple processable format. The edge node may receive some sensor data with missing values in the initial data processing. The improper data for predictive analysis may lead to incorrect results. Due to the processing constrained at the edge node, we have implemented the data imputation at the fog node. The edge node aggregated all the sensor data using Base64 encoding and sent it to the fog node. The received data at the fog node is decoded through the Base64 decoder and utilized for predictive analysis. Before data is used for predictive analysis, the fog node verifies the data's correctness. To detect the correctness of the data, the fog node process the data for missing sensor value. If data has a missing sensor value, the fog node processes the data to identify the correlation between the historical and current data. Since the data acquisition is scheduled through the RTC clock, it is much easier to correlate the current data with historical data of similar timestamps. The other non-missing neighbouring sensor feed in current data might correlate with historical data. We have implemented the Kd tree based on weighted variances and Euclidean distances to impute the missing sensor feed. The Kd tree searches the nearest neighbour of similar timestamps from the historical n proximity sensors and imputes the missing sensor feed by reading the nearest neighbours' sensor feeds. Later, the sensor data is preprocessed for anomalies detection. The local outliers factor (LOF) removes the anomalies from the dataset. The LOF technique finds the anomalous data point using the local deviation from its neighbors. The LOF score determines whether the data point is outliers or not.

$$\begin{cases} LOF(k) \sim 1 \ data \ point \ in \ a \ same \ cluster \\ LOF(k) < 1 \ data \ point \ is \ Inlier \\ LOF(k) > 1 \ data \ point \ is \ outlinear \end{cases}$$

In the above equation, k represents the locality with respect to its kth neighbors. To know the effectiveness of LOF, we have augmented the outlier data point in the dataset. The experimental result shows that LOF effectively detects the outlier from the data, as shown in Figures. 13 and 14.

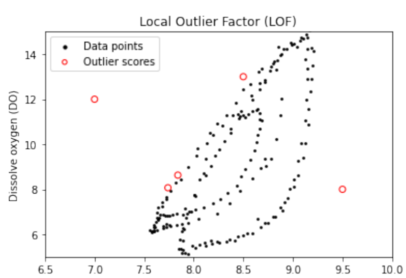

Fig. 13. Outlier detection between DO and pH

C. Aquaculture Control System

The actuators operated based on the control generated from the data analytics algorithm. The analytics algorithm is trained with historical data to predict the change in DO with relative water parameters temperature and pH. The system's sole dependence on sensor data may result in inaccurate monitoring and maintenance. Usually, the integrated system sensor's prolonged use introduces incorrect measurements. Despite the sensor's faulty measurement, we prevent the system's ideal state using the relative water parameters. The sensor measurement data is compared with historical sensor data of a similar timestamp. For DO prediction, we have water temperature, EC, and relative pH parameters. Similarly, we have considered the DO, EC and water temperature relative parameters for pH prediction. To analyze the correctness of the DO sensor data, we locate the pH and water temperature in historical data of similar time instances. To optimize the search, we have used a K-D tree that finds the nearest neighbour in a low dimensional real-valued space, as shown in Fig. 15.

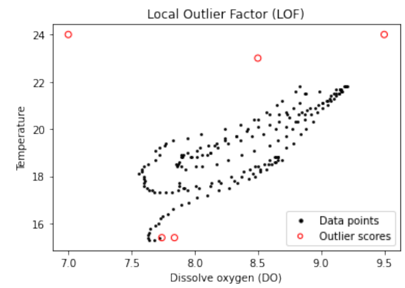

Fig. 14. Outlier detection between DO and Temperature

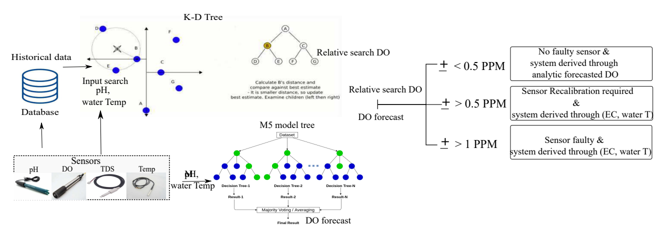

Fig. 15. Aquaculture control system

Similar data is substituted in the analytics algorithm for DO prediction. The predicted DO and relative search of DO from historical data may or may not have a near approximate value. If the relative difference is below 0.5 PPM between the predicted DO and search DO, then the system has no faulty sensor and schedule operation derived through an analytics algorithm. However, if the relative difference is more than 0.5 PPM, system sensors need recalibration and scheduled management derived through other relative parameters (EC and water temperature). To make an accurate prediction, we have considered robust, inexpensive multiple temperature probes and EC sensors as the relative parameter for DO and pH relative prediction. Similarly, the pH and DO are related to plankton algae photosynthesis, aquatic respiration, water temperature, and oxidative decomposition of organic matter. The change in pH is correlated due to ionic change in the water, which is easily inferred from the EC sensor. If the pH sensor shows inaccurate measurement, we stabilise the DO and pH relative change through the EC sensor. Similarly, if the relative difference is more than 1 PPM, then the system report relative sensor is faulty, and the system is derived through

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other relative parameters (EC and water temperature). The more relative parameters in combination give better accuracy for prediction. In this preliminary study, we have considered the pH, DO and EC, and water temperature relative parameters for analysing the water quality. Due to multiple relative sensor inputs, we have obtained the desired forecast from the analytics algorithm.

The integrated actuator is derived based on the fluctuation in the desired water parameters, as given in Table. I. The collected data scheduled the various actuator operations using RTC. Overnight excreted by-product and residual uneaten feed contributes to pH and DO change in the water, which is settled down to normal through RAS and aerator system with 6 am scheduled operation. Later at 10 am smart feeder system pump the plankton algae mixture into the harvesting tank and operates the aerator to supplement desired to DO requirement. The third sensor measurement time was set to 3 pm when mussels excreted by-product, and temperature variation may fluctuate the DO and pH. To balance the DO and pH, the proposed system, based on the requirement, may operate the RAS system and aerator. At 10 pm, scheduled actuator operation, we balance the DO and pH through RAS and aerator. Similarly, at the 1 pm overnight schedule, the proposed system analyses the DO requirement and accordingly operates the aerator. We simulate the system behaviour by changing the water parameters to validate the control system. The intelligent data analytics algorithm schedules the aquaculture management requirement based on the water parameters.

D. Performance evaluation for predictive analysis

The preprocessed data is used for predictive analysis. Initially, we calculated the correlation between the water parameters using bilateral Pearson correlation. Later, the high correlated data elements are combined for predictive analysis. A bilateral Pearson correlation is a type of linear correlation coefficient which returns the values between -1 to +1. A positive correlation means a strong positive correlation between the respective parameters, and a negative one means it is negatively linearly correlated. The correlation analysis results are shown in Table IV. The analysis shows that dissolved

TABLE IV
PEARSON CORRELATION ANALYSIS RESULTS.

Items	pH	EC	DO	Temp
pH	1.000000	0.192692	0.730997	0.828218
EC	0.192692	1.000000	0.003895	0.348716
DO	0.730997	0.003895	1.000000	0.521735
Temp	0.828218	0.348716	0.521735	1.000000

oxygen is highly correlated with temperature with correlation coefficients of -0.52. Similarly, pH is correlated with dissolved oxygen and temperature with correlation coefficients of 0.73 and 0.82, respectively. Because dissolved oxygen and Ph are the vital indicators for any aquaculture, we have considered these indicators in the training dataset. Similarly, water temperature significantly impacts chemical and biological reactions. We have included the water temperature also in our feature

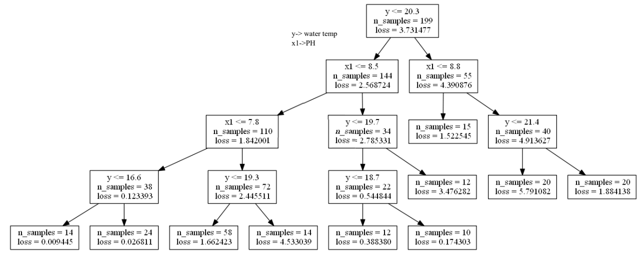

Fig. 16. Generated model tree with maximum depth=4

vector. Due to the high correlation between the DO, PH, and water temperature, we have trained the model by considering the parameter pH and temperature as the independent variable and dissolved oxygen as the dependent. We analyse the performance of tree-based models with the ANN model. We used the multilayer perceptron with ReLu as the activation function for the ANN model. Due to limited test measurement from the aquaculture site, we have kept 1000 validation runs using a training set of 160 samples and a test set of 40 samples to obtain the convergence of mean and standard deviation on the performance indicators. In M5 model tree construction, we have used the parameters such as linear model, maximum depth of decision tree = 4, the minimum number of samples at leaf=10, and greedy search strategy. The model tree training generates the ten rules, as shown in Fig. 16. For DO prediction, the M5 model tree shows a strong correlation R=0.877 between the actual and predicted data. The model estimated mean absolute error (MAE) was 0.963. Table.V shows the comparative performance of the M5 model tree for DO prediction with other non-linear decision tree-based predictors. The models are trained from 80 % of the sample (160). The remaining 20 % of the sample (40) is used to test the model performance. The non-linear RF, GBM, and M5 model tree performance are evaluated with the maximum tree depth of 4 and sample size of 40. All the non-linear tree-based model RF, GBM, and M5 model trees showed better performance compared to ANN. The slightly lower performance of ANN is due to the smaller size of the training dataset with multiple input parameter The comparative

TABLE V
RESULT OF DO PREDICTED FROM NON-LINEAR FORECASTING MODELS

Algorithms	Mean of real DO (mg/L)	Predicted Values		
		Correlation (R)	Mean (mg/L)	MAE
RF	9.121	0.7364	9.09	1.45
M5	9.121	0.8775	9.12	0.963
ANN	9.121	0.8578	9.10	1.463
GBM	9.121	0.6036	9.37	1.66

performance analysis obtained the lowest MAE of 0.963 from the M5 model tree. The model performance is validated using the new data set from the sensors. Similarly, we have evaluated the model performance for Ph prediction using a training sample containing DO, PH, and Temperature water parameters. Table. VI shows the comparative performance of the model tree for PH prediction with other non-linear decision tree-based predictors.

TABLE VI
RESULT OF PH PREDICTED FROM NON-LINEAR FORECASTING MODELS

Algorithms	Mean of real PH (ppm)	Predicted Values		
		Correlation (R)	Mean (mg/L)	MAE
RF	8.36	0.912	8.37	0.141
M5	8.36	0.974	8.362	0.01239
ANN	8.36	0.910	8.358	0.143
GBM	8.36	0.9299	8.37	0.118

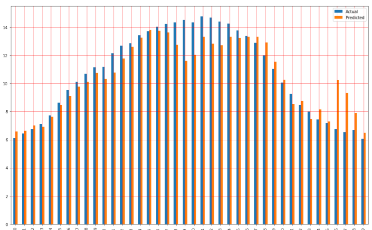

Fig. 17. DO Correlation between Actual vs Predicted

In Fig. 17 and Fig. 18, we have shown the M5 model tree correlation result between the actual and predicted data of PH and DO. The correlation between the actual and predicted values of DO was 0.8775 with an MAE of 0.963, which is quite a good prediction from the model. Similarly, the correlation result obtained for PH was 0.974 with an MAE of 0.01239. The experimental evaluation shows that the M5 model tree algorithm can predict the changing water parameters better than other RF, ANN, and GBM non-linear predictors. In the comparative study, we found M5 model tree is slightly better than other non-linear RF, GBM, and ANN models. Despite less training data, we obtain a better result with the M5 model tree due to its inherent features of grouping the high correlations data for constructing the M5-model tree.

The employed M5 model tree performs the group analyses of relative water parameters. We adopt the pruning methods for the model to generalize well to unseen data. The pruning method optimizes the tree structure without affecting the classification accuracy. The REP algorithm pruned the growing tree and used the information gained as the branching criteria. Redundant subtrees were pruned to solve the over-fitting problem and maximize the forecasting accuracy.

VI. CONCLUSION

In this paper, we have proposed a comprehensive IoT system for monitoring and maintenance of aquaculture. The proposed system design has a physical, network, and logical design. The physical design provided the hardware configuration details. In contrast, the network design shows the communication network between the physical devices. In this design, we have addressed the requirement of a fog node for intelligent data analytics and control. In the logical design, we have discussed the front-end and back-end implementation details. It also highlighted the importance of container-based virtualization and the various design features. This paper also discusses the importance of the core intelligent analytics algorithm for monitoring and managing aquaculture. The closed-loop control system ran through intelligent non-linear decision tree-based

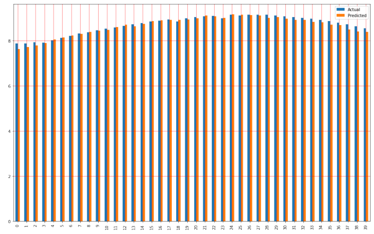

Fig. 18. PH Correlation (R) between Actual vs Predicted

models. This paper shows the comparative performance of the M5 model tree, RF, ANN, and GBM. The experimental evaluation found that the M5 model tree has the highest prediction accuracy for DO prediction with a correlation of 0.877 with an MAE of 0.963. Similarly, the M5 model tree outperforms PH prediction with a correlation of 0.975 and an MAE of 0.0123.

Future research in this direction includes more robust relative water sensors to make the system more affordable and accurate. In future research, we also include analysing and developing a more accurate forecasting model.

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